Research Brief

Difficulty-Stratified Evaluation of Confidence Calibration in Large Language Models

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Abstract

Large language models (LLMs) are increasingly deployed as decision-support systems, yet the fluency of their responses can mask substantial uncertainty. We evaluate Claude Haiku (Anthropic), GPT-4o-mini (OpenAI), and Gemini 3.1 Flash-Lite (Google) on an expanded, difficulty-stratified Massive Multitask Language Understanding (MMLU) bank of 540 multiple-choice questions (original 180 frozen items plus a 360-item content-disjoint supplement) spanning STEM, Social Sciences, and Humanities at easy, medium, and hard tiers fixed prior to evaluation. Each model completed five trials at temperature 0.3 (max 1024 output tokens) and provided both an answer and confidence estimate (0–100). With format-repair retries, parse success was 100% (8,100/8,100). Calibration was evaluated using accuracy, overconfidence gap, Expected Calibration Error (ECE), bootstrap 95% confidence intervals, and response-level difficulty regressions with Holm–Bonferroni correction. All models exhibited increasing overconfidence as difficulty increased, consistent directionally with the hard–easy effect in human judgment. Claude Haiku achieved the strongest calibration (89.5% accuracy; ECE $= 0.033$) with slight overall underconfidence, while GPT-4o-mini (71.5% accuracy; ECE $= 0.157$) and Gemini Flash-Lite (87.6% accuracy; ECE $= 0.117$) were overconfident, especially on hard items. Gemini additionally showed confidence collapse (95.6% of scores ≥ 95). For GPT-4o-mini, verbal confidence correlated only weakly with answer-token probability ($r = 0.22$). Verbal confidence is therefore an imperfect reliability signal and should complement, not replace, verification and human oversight.

Keywords Large Language Models (LLMs), Confidence Calibration, Task Difficulty, MMLU, Expected Calibration Error (ECE), Hard–Easy Effect, GPT-4o-mini, Gemini, Claude Haiku.

1. BACKGROUND AND MOTIVATION

Large language models (LLMs) are increasingly used as study aids, productivity assistants, and informal decision-support tools. Their responses are often fluent and delivered with an authoritative tone, which can encourage users to treat them as reliable even when the underlying information is incorrect or uncertain. Whether an LLM's expressed confidence accurately reflects its likelihood of being correct is therefore an important consideration for trustworthy AI.

Confidence calibration refers to the degree to which reported confidence aligns with observed accuracy. In cognitive psychology, confidence systematically varies with task difficulty: people tend to overestimate performance on difficult tasks and underestimate it on easier ones, the *hard–easy effect* (Moore & Healy, 2008), and limited competence is often accompanied by limited insight into one's own performance (Kruger & Dunning, 1999). In machine learning, Guo et al. (2017) showed that modern neural networks are frequently miscalibrated and introduced Expected Calibration Error (ECE), now a standard metric. Unlike traditional models, LLMs can express confidence directly in natural language, allowing calibration to be evaluated from verbally reported confidence rather than internal probabilities (Kadavath et al., 2022; Xiong et al., 2023).

Prior work established that strong predictive performance does not imply well-calibrated confidence, particularly under challenging conditions: pretrained Transformers calibrate poorly out of domain (Desai & Durrett, 2020), and generative QA models are frequently overconfident when wrong (Jiang et al., 2021). Lin et al. (2022) showed that models can produce verbal confidence estimates that reasonably track accuracy, while sampling-based alternatives such as semantic entropy (Kuhn et al., 2023; Farquhar et al., 2024) estimate uncertainty from multiple generations rather than the single self-report a user typically sees. Yin et al. (2023) found that LLMs remain substantially worse than humans at recognizing unanswerable questions, and Steyvers et al. (2025)

showed that people frequently overestimate LLM accuracy, so miscommunicated confidence carries real human–AI interaction costs.

Gap. Comparatively little student-accessible work has examined how verbally reported confidence changes as task difficulty systematically increases across a standardized knowledge benchmark, under a locked multi-model API protocol with response-level inference and provider-specific internal-probability availability. This brief addresses that gap.

2. RESEARCH QUESTION AND OBJECTIVE

Research question. How does task difficulty influence the calibration of verbally reported confidence in contemporary API language models, and does miscalibration grow as questions become harder?

Objective. Evaluate Claude Haiku, GPT-4o-mini, and Gemini 3.1 Flash-Lite on a difficulty-stratified MMLU bank under a common answer-plus-confidence protocol, quantifying calibration with accuracy, overconfidence gap, ECE, bootstrap intervals, and Holm-corrected response-level regressions of overconfidence on difficulty; and, where available (GPT-4o-mini), compare verbal confidence with answer-token log-probabilities.

3. METHOD

Experimental design. Three API language models—Claude Haiku, GPT-4o-mini, and Gemini 3.1 Flash-Lite—were presented with the same frozen, expanded subset of multiple-choice MMLU questions and required to provide both an answer and an explicit confidence estimate. Calibration was evaluated overall and across difficulty tiers fixed prior to evaluation.

Question bank. Questions were drawn from the MMLU test split (Hendrycks et al., 2021; Hugging Face `cais/mmlu`), restricted to STEM, Social Sciences, and Humanities (`config/domains.yaml`). Each subject was assigned to easy, medium, or hard using a mapping fixed prior to evaluation (`config/difficulty_map.yaml`), based on published MMLU subject-level performance rather than the evaluated models' performance. The bank comprises 540 items: (i) an original 180-item set sampled with seed 42 (up to 20 questions per domain×difficulty cell), frozen byte-for-byte under `data/v1_frozen/`; and (ii) a 360-item content-disjoint supplement sampled with seed 43 (`data/v2/bank_supplement_360.json`); combined as `data/v2/bank_full_v2.json`. One true duplicate pair in the original bank (`q0060`/`q0073`) is retained for archival comparability and noted in Limitations.

Evaluated models and inference. The three models were run under matched decoding conditions (Table 1):

Table 1. Language models evaluated in this study.

Model	Access Method	Identifier
Claude Haiku	Anthropic API	`claude-haiku-4-5-20251001`
GPT-4o-mini	OpenAI API	`gpt-4o-mini`
Gemini Flash-Lite	Google Gemini API	`gemini-3.1-flash-lite`

Table 2. Response parse rates by model.

Model	Parseable Responses
Claude Haiku	2700 / 2700 (100.0%)
GPT-4o-mini	2700 / 2700 (100.0%)
Gemini Flash-Lite	2700 / 2700 (100.0%)

temperature 0.3, maximum 1024 output tokens, five independent trials per question, for 8,100 planned responses ($540 \times 3 \times 5$). GPT-4o-mini requests additionally collected token log-probabilities for the answer letter. Open-weight local models (Llama 3.1, Mistral) were not collected in this submission. Source code: GitHub (v2-phase2-clients).

Confidence elicitation and parsing. Every inference call used the same structured prompt containing the question, four labeled choices, formatting instructions, and a valid response example, instructing the model to return only:

```
ANSWER: [A/B/C/D]
CONFIDENCE: [0-100]
```

This constrained protocol minimized formatting variability and avoided chain-of-thought as a confound. The primary prompt is locked character-for-character from the earlier protocol; on parse failure, up to two repair retries append a short format-repair footer without rewriting the locked template. With repair retries, parse success was 100% for all models (Table 2), so all 8,100 responses entered the analyses.

Metrics and statistical analysis. Correctness was scored against the official MMLU key; confidence was normalized to $[0, 1]$. Primary metrics were accuracy, mean confidence, overconfidence gap (mean confidence $-$ accuracy), and ECE with ten equally spaced bins. Difficulty dependence was quantified by response-level ordinary least squares regression of the per-response overconfidence gap on numerically coded difficulty (easy $= 1$, medium $= 2$, hard $= 3$); domain-specific ECE was a secondary analysis. For GPT-4o-mini, Pearson correlation between verbal confidence and answer-token probability was computed, along with ECE under each confidence source. Bootstrap 95% confidence intervals (2,000 resamples) were computed for accuracy, gap, and ECE. Six predeclared tests (slope $\neq 0$; hard-minus-easy gap > 0; per model) were corrected jointly with Holm–Bonferroni. Human hard–easy findings served only as a qualitative comparison. Configurations, banks, processed results (`data/v2/processed/`), and figure scripts are in the repository; analysis was produced by `src/analyze_v2.py` on collection stamp `20260716T175808Z`.

Table 3. Overall performance metrics by model (bootstrap 95% CIs).

Model	N	Accuracy	Mean Conf.	Gap	ECE
Claude Haiku	2700	0.895 [0.883, 0.907]	0.878	−0.017 [−0.027, −0.006]	0.033 [0.023, 0.044]
GPT-4o-mini	2700	0.715 [0.697, 0.731]	0.871	0.156 [0.140, 0.173]	0.157 [0.141, 0.175]
Gemini Flash-Lite	2700	0.876 [0.863, 0.889]	0.992	0.115 [0.104, 0.129]	0.117 [0.104, 0.129]

Table 4. Performance metrics by model and task difficulty.

Model	Difficulty	N	Accuracy	Mean Conf.	Gap	ECE
Claude Haiku	Easy	900	0.926	0.885	−0.041	0.043
Claude Haiku	Medium	900	0.890	0.869	−0.021	0.045
Claude Haiku	Hard	900	0.869	0.880	0.011	0.037
GPT-4o-mini	Easy	900	0.847	0.874	0.028	0.036
GPT-4o-mini	Medium	900	0.783	0.870	0.086	0.088
GPT-4o-mini	Hard	900	0.514	0.868	0.354	0.354
Gemini Flash-Lite	Easy	900	0.949	0.990	0.041	0.045
Gemini Flash-Lite	Medium	900	0.876	0.992	0.117	0.117
Gemini Flash-Lite	Hard	900	0.804	0.993	0.189	0.189

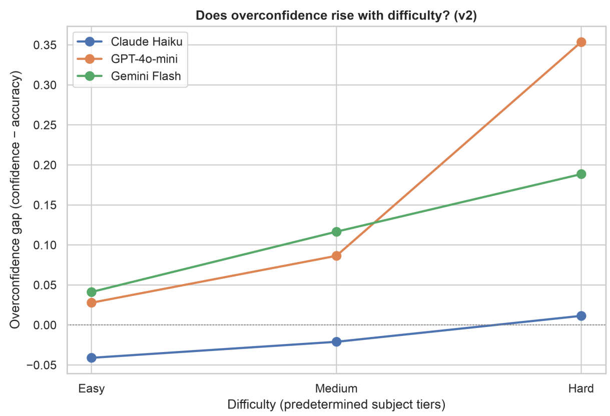

Figure 1. Overconfidence gap across task difficulty levels (v2; predetermined subject tiers).

4. EVIDENCE AND RESULTS

All analyses include the full set of 8,100 successfully parsed responses.

4.1. Overall performance

Table 3 summarizes accuracy, mean confidence, overconfidence gap, and ECE. Claude Haiku achieved the highest accuracy with the lowest calibration error and slight aggregate underconfidence. GPT-4o-mini showed the lowest accuracy and largest ECE. Gemini Flash-Lite was highly accurate but substantially overconfident because mean confidence remained near ceiling (0.992).

4.2. Calibration across task difficulty

Table 4 and Figure 1 show calibration across difficulty. For all three models, the overconfidence gap rises from easy to hard. The increase is steepest for GPT-4o-mini (gap $0.028 \rightarrow 0.354$), driven by a large accuracy drop on hard items while mean confidence remains nearly flat. Gemini's gap also rises ($0.041 \rightarrow 0.189$) despite near-ceiling confidence at every tier. Claude remains closest to zero throughout, moving from mild underconfidence on easy items to slight overconfidence on hard items.

Table 5. Confidence dispersion by model (0–100 scale).

Model	Mean	SD	% $\geq$ 95
Claude Haiku	87.8	13.4	42.6%
GPT-4o-mini	87.1	3.6	4.9%
Gemini Flash-Lite	99.2	2.5	95.6%

Table 6. Expected calibration error by model and domain.

Model	Domain	ECE	N
Claude Haiku	Humanities	0.047	900
Claude Haiku	STEM	0.012	900
Claude Haiku	Social Science	0.057	900
GPT-4o-mini	Humanities	0.156	900
GPT-4o-mini	STEM	0.247	900
GPT-4o-mini	Social Science	0.087	900
Gemini Flash-Lite	Humanities	0.128	900
Gemini Flash-Lite	STEM	0.141	900
Gemini Flash-Lite	Social Science	0.081	900

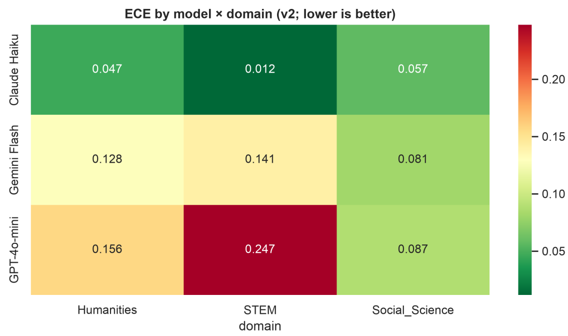

Figure 2. ECE heatmap by model and domain (v2; lower is better).

4.3. Confidence dispersion

Gemini Flash-Lite exhibits *confidence collapse*: near-ceiling mean confidence with very low dispersion (Table 5). Rising overconfidence for Gemini is therefore driven primarily by falling accuracy rather than informative modulation of stated confidence.

4.4. Domain-level performance

Claude Haiku maintains low ECE across domains (Table 6; Figure 2). GPT-4o-mini shows the largest domain ECE on STEM (0.247). Gemini is intermediate across domains.

4.5. Reliability diagrams and capability scatter

Figure 3 shows reliability diagrams. Claude Haiku remains closest to the ideal diagonal; GPT-4o-mini and Gemini deviate more, consistent with higher ECE. Figure 4 summarizes the capability–calibration relationship in this three-model sample: higher accuracy tracks lower ECE, with Claude best on both axes and GPT-4o-mini worst.

4.6. Response-level difficulty regression

Unlike three-point cell-mean regressions, Table 7 uses all responses. Difficulty significantly predicted

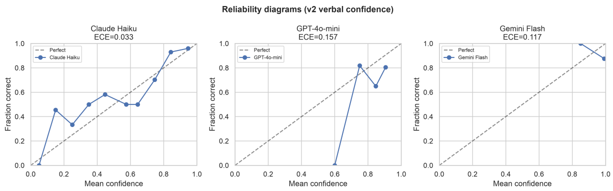

Figure 3. Reliability diagrams comparing verbal confidence calibration (v2).

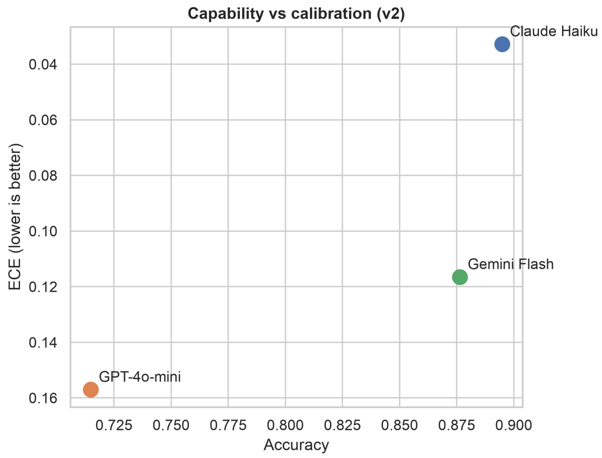

Figure 4. Accuracy versus ECE across evaluated models (v2).

Table 7. Response-level OLS of per-response overconfidence gap on difficulty (easy=1, medium=2, hard=3), with Holm-corrected p-values.

Model	N	Slope	SE	95% CI	p (Holm)	R^2
Claude Haiku	2700	0.026	0.007	[0.013, 0.039]	7.8×10^{-5}	0.006
GPT-4o-mini	2700	0.163	0.010	[0.143, 0.183]	3.1×10^{-56}	0.089
Gemini Flash-Lite	2700	0.074	0.008	[0.059, 0.088]	7.3×10^{-22}	0.034

overconfidence for all three models after Holm correction. Effect sizes are modest for Claude ($R^2 = 0.006$) and larger for GPT-4o-mini ($R^2 = 0.089$). Hard-minus-easy gap contrasts were likewise significant after Holm correction for every model.

4.7. Verbal versus internal confidence (GPT-4o-mini)

For GPT-4o-mini, verbal confidence and answer-token probability are positively but weakly correlated (Table 8). ECE using internal answer probability is *worse* than ECE using verbal confidence on this protocol. Claude and Gemini are excluded because Anthropic's public API does not expose token log-probabilities, and Gemini returned only aggregate `avgLogprobs` rather than per-token answer probabilities.

4.8. Robustness: original 180 versus full 540

Direction and ranking of calibration are stable under bank expansion (Table 9).

Table 8. Verbal versus answer-token internal probability (GPT-4o-mini only).

Metric	Value
N with internal prob.	2700
Pearson r (verbal, internal)	0.222
ECE (verbal)	0.157
ECE (internal answer prob.)	0.243

Table 9. Original-180 subset versus full-540 bank (accuracy / gap / ECE).

Bank	Model	Acc.	Gap	ECE
original_180	Claude Haiku	0.911	−0.020	0.037
full_540	Claude Haiku	0.895	−0.017	0.033
original_180	GPT-4o-mini	0.726	0.148	0.152
full_540	GPT-4o-mini	0.715	0.156	0.157
original_180	Gemini Flash-Lite	0.874	0.117	0.119
full_540	Gemini Flash-Lite	0.876	0.115	0.117

5. DISCUSSION

The primary finding is that all three evaluated models exhibit increased overconfidence as task difficulty increases (Table 4; Figure 1). Although accuracy declines on harder questions, reported confidence does not decrease proportionally, producing larger confidence–accuracy discrepancies precisely where performance is weakest. Claude Haiku follows the same directional trend but with much smaller magnitude, remaining near-calibrated overall (ECE $= 0.033$; gap $= -0.017$).

Substantial differences emerge across models. Claude combines high accuracy (0.895) with the best calibration. GPT-4o-mini has lower accuracy (0.715) and the steepest response-level difficulty slope (0.163), with hard-tier gap reaching 0.354. Gemini is highly accurate (0.876) yet overconfident (gap $=$ 0.115; ECE $=$ 0.117) because confidence remains near ceiling with very low dispersion—a pattern better described as confidence collapse than as informative metacognitive modulation (Table 5).

Because all three arms are API models, we do *not* attribute calibration differences to open-weight versus proprietary architecture. In this sample, calibration tracks capability (Figure 4), with provider and model class confounded with accuracy. The observed difficulty–overconfidence relationship is directionally consistent with human hard–easy research (Moore & Healy, 2008) but should not be interpreted as evidence that LLMs share human metacognitive mechanisms (Kruger & Dunning, 1999); the empirical parallel is narrower: as evaluation difficulty increases, model confidence becomes less aligned with correctness.

The GPT-4o-mini verbal-versus-internal comparison further cautions against treating either signal as a complete reliability measure: the two are only weakly correlated, and internal answer-token probabilities were not better calibrated than verbal reports under this elicitation protocol (Table 8). Practically, a reported confidence score is attractive but model-dependent. For Claude Haiku, verbal confidence is comparatively informative. For GPT-4o-mini on hard items and for Gemini's near-ceiling scores, high confidence frequently co-occurs with error. Confidence estimates should therefore complement external verification, especially on difficult STEM tasks.

6. LIMITATIONS

Evaluation is restricted to multiple-choice MMLU items; findings may not generalize to open-ended generation or tool-augmented agents. Difficulty is assigned via a subject-level mapping fixed prior to evaluation rather than item-level IRT. The original 180-item bank contains one true duplicate pair (`q0060`/`q0073`).

Confidence is primarily verbal (0–100). Internal answer-token probabilities were available only for GPT-4o-mini; Claude and Gemini are excluded from that comparison by API design, not by silent data loss. Constrained `ANSWER`/`CONFIDENCE` formatting suppresses chain-of-thought and may affect absolute accuracy relative to CoT-style leaderboards; claims concern miscalibration under this protocol.

Open-weight local models were not collected in this submission. No formal human-subjects baseline is included. Results are specific to temperature 0.3, five trials, max 1024 output tokens, the 540-item bank, and July 2026 model versions.

7. CONCLUSION AND NEXT STEPS

Under a locked verbal confidence protocol on an expanded 540-item MMLU bank with five trials per item, Claude Haiku, GPT-4o-mini, and Gemini 3.1 Flash-Lite all show rising overconfidence with predetermined difficulty. Claude remains near-calibrated; GPT-4o-mini shows the steepest difficulty-linked overconfidence; Gemini combines high accuracy with confidence collapse. Verbal confidence is therefore not a universally reliable indicator of correctness and should be treated as a supporting signal rather than a substitute for verification. Calibration must be evaluated where models are most likely to fail—on hard items—not only where average performance appears strong.

Next steps. Future work should extend to MMLU-Pro, GPQA, and related harder benchmarks; reintroduce open-weight local models under the same locked protocol; expand item-level difficulty estimation; systematically compare elicitation strategies; and connect calibration metrics to human reliance behavior. Where providers expose token probabilities, verbal-versus-internal comparisons should be repeated across model families.

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AI USE TRANSPARENCY STATEMENT

AI tools were used in this project in several supporting roles. The language model APIs evaluated in this study (Anthropic Claude Haiku, OpenAI GPT-4o-mini, and Google Gemini Flash-Lite) were the *subjects* of the experiment; all inference calls, prompts, and raw outputs are archived in the project repository. Cursor Agent was used as a software development assistant to help generate and refactor code, navigate the project structure, suggest terminal commands, assist with debugging, and support dataset extraction and file organization. ChatGPT and Claude were used to assist with LaTeX formatting, improve the technical clarity and organization of the manuscript, and refine the language of the paper without altering the underlying methodology, analyses, or conclusions. The software used for data collection, processing, statistical analyses, figure generation, and table generation was developed by the authors with assistance from Cursor Agent. The study design, implementation, interpretation of results, and final manuscript content were constructed and reviewed by the authors, who verified all experimental results, calculations, figures, tables, claims, and references against the original sources.